

Assessing Performance at the Age of Artificial Intelligence and Digital Theatre in Africa

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Abstract

Africa stands at a historic crossroads between tradition and innovation, between the organic pulse of ritual performance and the algorithmic rhythms of artificial intelligence. As the world's theatres increasingly integrate AI-driven dramaturgy, virtual scenography, and interactive digital environments, African performance practices are compelled to negotiate their identity within this global shift. This paper examines the implications of AI and digital theatre on African performance, situating the discussion within postmodern and posthuman frameworks. It interrogates Africa's technological readiness, cultural responses, and aesthetic transformations in an age when humans and machines co-create meaning. Drawing upon descriptive-analytical observation, the paper explores global and African discourses on digital dramaturgy, highlighting both infrastructural limitations and creative opportunities. It concludes that Africa's future in the digital humanities depends not merely on adopting technology but on Africanizing it, embedding AI within the continent's performative cosmology to achieve a decolonised digital theatre.

Keywords: Artificial Intelligence, Digital Theatre, Postmodernism, Posthuman Performance, African Theatre, Devised Creation

Introduction

The twenty-first century has ushered in a profound redefinition of creativity. Performance, once bound by the physicality of the stage, now thrives in virtual, augmented, and algorithmic spaces. The advent of Artificial Intelligence (AI) has expanded the boundaries of imagination, introducing machine learning as a co-creator in art, literature, and theatre. Global performance practices increasingly embrace AI not merely as a tool but as a collaborator capable of reconfiguring authorship, embodiment, and audience interaction.

Within this expanding digital ecosystem, African theatre faces a complex duality: the desire to preserve its communal, ritual-centred essence while confronting the inevitability of technological evolution. African theatre has always been performative beyond the proscenium, rooted in social participation and symbolic exchange. However, the arrival of digital technology challenges this continuity, demanding a shift from oral-based dramaturgy to algorithmic aesthetics. Traditional African Theatre relies heavily on physical presence, communal participation, and oral storytelling. However, the rapid rise of AI and digital technologies is redefining these paradigms. Virtual reality, livestreaming and algorithmic

creation are creating new modes of artistic expression.

has emphasised that Africa's performance traditions are inherently adaptive; they absorb innovation while preserving cultural identity. However, the continent's uneven access to digital infrastructure and cautious engagement with AI-driven art reveal deep-seated anxieties about cultural dilution, creative autonomy, and epistemic control.

This paper, therefore, seeks to assess Africa's readiness to embrace AI and digital theatre as an extension, not a replacement of its aesthetic and philosophical foundations. Also, this paper proposes a localised framework for assessing theatrical performance in the digital age. It adopts a postmodern and posthuman framework to interrogate how African performance can reinvent itself through digital hybridity while remaining grounded in indigenous cosmologies of storytelling, rhythm, and communal embodiment.

This paper seeks to bridge the gap between indigenous performance, traditions, and global technological advancement, offering fresh insights for theatre scholars, digital artists, and cultural policymakers.

Theoretical Framework: Postmodernism, Posthumanism, and the Digital Turn

Postmodernism emerged as a critique of universal truths and grand narratives, favouring multiplicity, hybridity, and intertextuality. It finds resonance in digital performance, which thrives on fragmentation, simulation, and shared authorship. In postmodern theatre, meaning is not fixed but generated through interactivity, a dynamic now mirrored by AI systems that can co-author narratives and generate performative responses in real time. . With this quote, Postmodernism rejects the fixed hegemony of the physical stage; rather, it accommodates digital theatre in Africa, such as Zoom plays, WhatsApp performances, and hybrid augmented-reality experiences. Pastiche shows us that African digital performance blends with indigenous oral traditions. These include the Yoruba Alaarinjo or Hausa Emisai theatre. Often, the blend comes with electronic music, green scenes, etc.

The notion of simulacra and hyperreality provides a useful lens for analysing digital performance in the AI age. The algorithmic stage becomes a mirror hall of simulations where avatars, projections, and virtual presences coexist with human performers. This collapse of the real and the virtual parallels African metaphysical traditions, where the visible and invisible worlds coexist in ritual performance. Thus, AI-mediated theatre is not entirely alien to African cosmology; it echoes the idea of multiple realities operating simultaneously.

Complementing postmodernism, posthumanism reframes the human as part of an interdependent network of machines, systems, and ecologies.. In this paradigm, creativity is no longer an exclusively human act but a distributed phenomenon, thereby making humans co-creators of art with machines. The theatre of the posthuman era, through algorithmic choreography, machine improvisation, or virtual dramaturgy, embodies this decentred authorship and also brings about the erosion of mimetic authenticity.

For African performance, posthumanism opens radical possibilities. It aligns with Ubuntu philosophy, the idea that "I am because we are," extending the collective identity of

performance to include nonhuman collaborators. The drum, mask, or ancestral spirit in traditional theatre already represents an agentive presence beyond the human. The inclusion of AI extends this ontology into the digital domain, allowing technology to become part of the African performative ensemble.

LITERATURE REVIEW

3.1 Global Perspectives on AI and Theatre

Across the global north, the past two decades have seen rapid experimentation with AI-assisted theatre. Projects such as *AI: When a Robot Writes a Play* and *SPILL: Machine Dramaturgy Project* reveal how machine learning can generate scripts, simulate emotional tone, and interact with human actors. In the play, rather than a traditional overarching linear narrative, the play is structured into 8 episodes showing how a robot navigates typical human milestones and emotions. However, in the play, the AI frequently struggled with gender consistency, sometimes mixing up which character was male or female. . . argues that AI dramatists now function as dramaturgical “assistants,” producing textual material that human artists curate to yield better output.

Similarly, introduce the concept of “posthuman dramaturgy,” where algorithmic unpredictability mirrors improvisation, a key theatrical impulse. Scholars such as have further noted that AI challenges the notion of creativity itself by decentralising authorship, forcing artists to confront the ethical and aesthetic implications of nonhuman collaboration.

In devised theatre, AI facilitates what calls “algorithmic co-authorship”: a process in which digital systems analyse emotional tone, generate alternate endings, or even suggest rhythm changes in performance. These innovations highlight AI's potential as both a creative partner and a cultural mirror.

3.2 African Theatre and Digital Performance

African scholarship, though limited in volume, has begun to engage with these developments. Balme (2008) frames African theatre within the “postcolonial public sphere,” emphasising the need to balance global innovation with local authenticity. Critiques how Western performance technologies can inadvertently reproduce colonial hierarchies of knowledge if not decolonised. This can be done through the deliberate development of algorithms or operators that recognise African-specific peculiarities in performance and aesthetics.

Recent studies ' Highlight how African artists used digital platforms during the COVID-19 pandemic to sustain performance communities through livestreamed plays, Zoom festivals, and social media storytelling. These experiments demonstrated resilience but also exposed infrastructural inequalities, especially in power supply, broadband access, and software localisation.

There is a growing awareness that Africa must define its own digital humanities framework. This means embedding technology within indigenous dramaturgies rather than importing aesthetics wholesale from the West. The challenge is not a lack of creativity but a lack of systems

integration that bridges artists, technologists, and institutions. This is because digital technologies are not neutral creative agents in African arts. They have been created with the separate languages of the advanced worlds.

3.3 Nigerian and Pan-African Scholarship

In Nigeria, universities and independent collectives have begun exploring digital dramaturgy through student-led performances that integrate projection mapping, AI-generated soundscapes, and hybrid improvisations. These works, often undocumented in mainstream academia, signal a silent revolution in African creative education.

Scholars such as and " propose that African theatre's survival depends on its ability to negotiate the interface between ritual and algorithm between ancestral symbolism and contemporary digital logic. This negotiation, they argue, must occur within African epistemological frameworks that honour performance as a site of communal knowledge rather than mere entertainment.

4. Methodology

This study employs a descriptive-analytical method, supported by theoretical interpretation. It examines existing literature, performance practices, and digital trends to assess Africa's readiness to integrate AI and digital theatre. The research draws from both qualitative observation of digital performance experiments and content analysis of academic sources.

The theoretical lens combines postmodernism (to address aesthetic hybridity) and posthumanism (to interpret AI's presence as a performative agent). Observation focused on case examples from African universities, online theatre festivals, and experimental projects documented between 2018 and 2024. The data analysis privileges philosophical interpretation and critical synthesis over empirical quantification.

5. Findings and Discussion

5.1 Africa's Technological Readiness

Africa's readiness for AI-driven performance remains aspirational. Despite creative potential, infrastructural limitations, power instability, low broadband coverage, and insufficient digital training, these impede large-scale innovation. However, the narrative is not entirely bleak. Countries such as Nigeria, Kenya, and South Africa have made significant advances through digital arts incubators and creative technology startups. Initiatives like the Nairobi Digital Lab and Lagos Theatre in Virtual Space demonstrate how small-scale projects can prototype hybrid performances integrating AI-assisted sound design and projection technologies.

This reflects what — terms the “localised innovation model,” where regions develop context-specific applications of global technologies. African theatre, therefore, must envision AI not as imported software but as adaptable craft, embedded in cultural consciousness.

5.2 Tradition versus Innovation: The Aesthetic Dialectic

The tension between preservation and transformation defines much of Africa's engagement with technology. Traditional theatre thrives on communal participation and ancestral invocation, qualities that appear antithetical to algorithmic precision. However, both share an improvisatory logic.

In indigenous Nigerian theatre, for instance, the drum orchestrates collective rhythm and dictates the tempo of storytelling. In AI dramaturgy, algorithms perform similar functions, analysing timing, sentiment, and rhythm in real time. The intersection reveals that both forms rely on pattern recognition: the drummer reads human emotion, while AI reads data patterns.

argues that African performance evolves through absorption rather than replacement. Likewise, AI can be assimilated into the role of a modern griot, a digital interpreter of memory. When AI tools are trained on African oral traditions, languages, and aesthetics, they can extend the lifespan of cultural memory rather than erode it.

5.3 Postmodern Aporia and Digital Consumerism

Baudrillard's aporia, the uncertainty of meaning in a hyperreal world, echoes Africa's cultural ambivalence toward digital technology. The stage has become a mediated spectacle, streamed, edited, and consumed across platforms. While this expands audience reach, it risks reducing content performance.

As " warns, digital consumerism commodifies theatre, privileging aesthetics over essence. However, this same mediation democratizes access to performance, allowing rural artists to connect globally. The challenge, therefore, is curatorial literacy, training African practitioners to use AI and digital platforms critically rather than passively.

5.4 Posthuman Dramaturgy and the Future of African Creativity

AI challenges traditional authorship. Machine-learning systems trained on textual or performative data can co-create scripts, analyse audience engagement, or autonomously design visual elements. In this scenario, the artist becomes a curator of possibility rather than a sole originator.

For African theatre, this decentering resonates with communal creation, the "we" rather than the "I." The performer, audience, and environment have always co-produced meaning. Integrating AI extends this circle of collaboration. The posthuman African artist, therefore, becomes both coder and griot, a custodian of stories that traverse human and machine consciousness.

Other Key Aspects to Pay Attention to on African Performance and Artificial Intelligence

This section further draws attention to the fact that AI is formed through an algorithm that does not capture the peculiarities of Africa in terms of its multidimensional cultures and peoples, and its peculiar sociology, which often shows these in the form of drama and theatre. Therefore, there is a need for more human effort to engage with AI when preparing for a performance than when an American is doing the same. This is because America owns numerous data centres and oversees all algorithms that determine the results produced by this sophisticated digital infrastructure. The 'The Influencer' in this sense ensures that artists involved in collaborative arts

of this nature do not fall out of tune with the sociocultural peculiarities of Africans. The following are the aspects:

1. Scriptwriting and Translation: In deploying generative AI for writing scripts, this catalyst that we have called an influencer must ensure that African folklore, orature and language or linguistic aspects are lost in the process of artistic creation.
2. In creating highly immersive environments suitable for performances of any nature that are African, whether for 3D rendering of set design or costumes conceptualisation, infuse nuances and meanings that are originally African. This will prevent alienation in such creations.
3. Acting and rehearsal using an AI-assisted package must not rob Africans of their peculiarities in terms of movements, folk-sounds, music, which should not function as an accompaniment in any way; these two activities must carry the spirit of the continent about its traditionality.
4. Also, audience engagement and accessibility are being taken to an unprecedented level using ai algorithms and mixed reality. This fosters robust integration among different races and cultures. However, this integration does not imply consensus on every front of the ongoing discourse and the contexts therein. Africa must protect its identity within the smorgasbord of prevailing digital revolution.

6. Recommendations

1. Digital Infrastructure Investment: African governments must prioritise the creative economy within digital policy frameworks. National theatres and universities should have access to software, hardware, and stable power for AI-assisted creation.
2. AI Literacy in Theatre Education: Curricula should integrate AI ethics, coding fundamentals, and creative technology training into theatre studies to bridge the gap between artistry and computation.
3. Indigenous Dataset Development: To prevent algorithmic bias, African linguists and technologists should collaborate in developing AI models trained on African languages, gestures, and aesthetics.
4. Cross-Disciplinary Collaborations: Partnerships between theatre departments and computer science faculties can foster joint research on virtual scenography, machine improvisation, and digital performance analysis.
5. Policy and Cultural Archiving: Governments should support digital preservation of indigenous performances to ensure that AI systems learn from authentic African epistemologies.
6. International Exchange: Establish regional networks and global collaborations with digital humanities labs in Europe, Asia, and the Americas for skill-sharing and co-productions.

7. Conclusion

Africa's artistic destiny in the age of AI will not be defined by imitation but by interpretation. To remain culturally relevant, African theatre must transform AI from an imported instrument into an indigenized ally, one that listens to the drumbeat of the continent's evolving consciousness.

This paper has shown that the intersection between African theatre and artificial intelligence is not merely technological but philosophical. It involves reclaiming the power to narrate one's own stories in the context of digital modernity. The challenge before African artists and scholars is to navigate this threshold with discernment, to embrace posthuman collaboration without surrendering human essence.

As digital theatre evolves globally, Africa's contribution will lie not in reproducing Western aesthetics but in exporting new dramaturgies grounded in rhythm, ritual, and communal memory. The future African stage, augmented by AI but anchored in spirit, will embody the fusion of tradition and technology, continuity and disruption, humanity and code.

One cannot conclude this paper properly without acknowledging the efforts that are being made by the Society of Nigerian Theatre Artists (SONTA) and Arts and Culture Trust in South Africa in ensuring that intellectuals meet and discuss profound issues and subtle underlying situations which African artists have had to resolve as they adopt/adapt AI in Africa's creative industry.

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